Testing Higher Doses of Sildenafil to Repair Brain Injury Secondary to Birth Asphyxia: An Open-Label Dose-Finding Phase 1b Clinical Trial–Sildenafil Administration to Treat Neonatal Encephalopathy–Study 02

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Objective To evaluate the safety and tolerability of higher doses of sildenafil in neonates with hypoxic-ischemic encephalopathy (HIE) and brain injury.

Study design A phase 1b open-label dose-finding clinical trial in neonates with moderate-severe HIE and confirmed brain injury on a day-2 magnetic resonance imaging during therapeutic hypothermia (TH). Enteral sildenafil was administered every 12 hours (q12 h) for 7 days. All participants received an initial dose 2.0 mg/kg, and a second dose of 2.5 mg/kg. Starting from the third dose, group 1 received 2.5 mg/kg q12 h and group 2 received 3.0 mg/kg q12 h. Primary outcome was incidence of dose-limiting toxicities. Secondary outcomes explored day-30 neuroimaging and 18-month neurodevelopment.

Results Among the 30 neonates born between October 2019 and December 2021, 20 displayed day-2 brain injury and 13 received sildenafil (8 in group 1; 5 in group 2). In group 1, 25% (2/8) experienced transient hypotension after the first dose, linked to antiseizure medications. No significant hypotension occurred in group 2 when sildenafil was administered separately. At the 3.0 mg/kg/dose, steady-state sildenafil concentrations persisted beyond TH. Death or significant 18-month neurodevelopmental impairment occurred in 50% (4/8) of group 1 and 60% (3/5) of group 2. Among the survivors, partial recovery of brain injury was seen in 80% (4/5) of group 1 and 75% (3/4) of group 2; cerebral palsy developed in 0% (0/5) and 50% (2/4), respectively.

Conclusions Enteral sildenafil up to 3.0 mg/kg q12 h was safe and well tolerated in a small single-center cohort of neonates with HIE treated with TH. Phase 2 trials are needed to assess multicenter feasibility and efficacy. *(J Pediatr 2025;285:114701)*.

Trial registration ClinicalTrials.gov NCT04169191.

Recently, sildenafil has been proposed as a potential neurorestorative treatment for neonates with hypoxic-ischemic encephalopathy (HIE) treated with therapeutic hypothermia (TH).[1-4] Preclinical rodents studies have demonstrated that sildenafil, alone or with TH, has neuroprotective and neurorestorative properties in the context of hypoxia-ischemia.[3-6] Proposed mechanisms include reducing apoptosis, promoting neurogenesis, decreasing neuroinflammation, and enhancing myelination.[1-5] In a recent phase 1b trial (Sildenafil Administration to treat Neonatal Encephalopathy – Study 01 [SANE-01], NCT02812433), we enrolled higher-risk neonates with HIE treated with TH and confirmed day-2 brain injury; we demonstrated that enteral sildenafil at a dose of 2.0 mg/kg/dose every 12 hours (q12 h) from day 2 of life to day 9 was safe and well-tolerated.[2]

Preclinical data have suggested that higher doses of sildenafil may be more effective for brain injury recovery.[3,4] Although 6 mg/kg/day of sildenafil is the

aEEG	Amplitude-integrated electroencephalogram	q12 h	Every 12 hours
BG	Basal ganglia	SAE	Serious adverse event
CP	Cerebral palsy	SANE-01	Sildenafil Administration to treat Neonatal Encephalopathy–Study 01
DMC	Data Monitoring Committee		
GDD	Global developmental delay	SANE-02	Sildenafil Administration to treat Neonatal Enephalopathy–Study 02
HIE	Hypoxic-ischemic encephalopathy		
MRI	Magnetic resonance imaging	TH	Therapeutic hypothermia

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most cited maximum tolerable dose for neonatal pulmonary hypertension,[7-11] its safety in neonates with HIE treated with TH remains unclear, as these neonates have typically been excluded from previous studies. Infants with HIE are at risk for complications such as hypotension, pulmonary hypertension, acute kidney injury, and liver impairment.[12-15] Determining the highest safe dose of sildenafil that offer neuroprotection or neurorestoration without compromising cardiopulmonary stability is essential. In SANE-01, a conservative dose of 2.0 mg/kg q12 h was used to limit exposure during TH.[2] This dose reached a steady-state concentration during TH, similar to level reported by Ashman et al[8] in neonates with pulmonary hypertension. However, this concentration declined rapidly after TH was completed.[2] Thus, our objective was to evaluate the tolerability and safety of higher sildenafil doses in neonates with moderate-severe HIE and early brain injury despite TH.

Methods

Trial Design

Sildenafil Administration to treat Neonatal Enephalopathy-Study 02 (SANE-02) was a phase 1b, open-label, two-arm, dose-finding clinical trial conducted at a single tertiary neonatal intensive care unit (NCT04169191). The trial enrolled neonates with HIE, who exhibited brain injury while undergoing TH. Neonates with HIE treated with TH born at ≥36 weeks and weighing ≥1800 g, with moderate-severe HIE indicated by amplitude-integrated electroencephalogram (aEEG) upon admission, and hypoxic-ischemic brain injury confirmed by day-2 (baseline, before study medication) brain magnetic resonance imaging (MRI) were eligible for the study.[2,16,17] Neonates who were moribund or had a genetic syndrome, complex congenital heart disease, cerebral malformations, or intraventricular or intraparenchymal hemorrhage were excluded. Before starting the study, we obtained Health Canada and local ethics board approvals. Enrollment occurred after written informed consent was provided by the neonates' legal representatives.

Each eligible neonate received enteral sildenafil twice daily for 7 consecutive days, totaling 14 doses from day 2 to day 9 of life. To test the tolerability and safety of higher doses, we used a dose-escalation design to test two dose groups.[18,19] In each dose group, at least three neonates with HIE treated with TH received all 14/14 doses of enteral sildenafil, allowing for a safety assessment before advancing to the next dose level. Group 1 received an initial dose of 2.0 mg/kg, followed by a second dose of 2.5 mg/kg, with all subsequent doses being 2.5 mg/kg q12 h (5.0 mg/kg/day). Group 2 also started with a 2.0 mg/kg dose followed by a second dose of 2.5 mg/kg; however, starting from the third dose, the dosage was increased to 3.0 mg/kg q12 h (6.0 mg/kg/day). An independent Data Monitoring Committee (DMC) reviewed serious adverse events (SAEs) and safety data after at least three neonates completed the full 14-dose treatment regimen. After enrolling at least three neonates with HIE treated with TH in group 2, corresponding to the described upper threshold of sildenafil,[7-11] a dose-expansion phase was planned to gather additional safety data. The sildenafil suspension was ordered from Gentes & Bolduc Pharmacists and had a concentration of 2.5 mg/mL, as outlined in the US Pharmacopeia (Sildenafil Compounded Oral Suspension, US Pharmacopeia 36-NF31, April 2013). Criteria for discontinuation included: (1) significant hemodynamic compromise defined as the need for two inotropic or vasopressor agents in addition to hydrocortisone for blood pressure support or the initiation of extracorporeal membrane oxygenation; (2) the detection of a new intraventricular or intraparenchymal cerebral hemorrhage; or (3) a parental decision to transition to palliative care. During the study period, neonates with HIE were kept nil per os throughout the entire duration of TH.

Data Collection

We collected data on gestational age, birth weight, sex, delivery mode, 10-minute Apgar score, arterial cord pH, initial blood gas pH, lactate levels at admission, and outcomes (survival and 18-month neurodevelopment).[2] The severity of HIE at admission was assessed using the modified Sarnat score and aEEG background activity.[2,20-23] We also recorded clinical signs of pulmonary hypertension (difference between the pre- and the postductal saturation of greater than 10%, associated with the use of inhaled nitric oxide). Baseline measurements of troponin I, creatine kinase, creatinine, aspartate aminotransferase, alanine aminotransferase, C-reactive protein, and white blood cell count were collected to evaluate the severity of multiorgan dysfunction and any concurrent infections at baseline, that is, prior to starting the study medication. An ophthalmologist performed a fundus exam on day 10 to assess retinal health.

Safety

Safety was evaluated by tracking AEs, including death, worsening hypotension, progression of clinical pulmonary hypertension, liver dysfunction, intraventricular/intraparenchymal hemorrhage, or retinopathy.[2] The categorization of AEs was predetermined based on previously published studies.[2,24-28] Hypotension was classified as a SAE if the mean blood pressure dropped by >10 mmHg or to an absolute value <35 mmHg within 4 hours of administering the study medication.[11] For each dose of sildenafil, vital signs were recorded every 15 minutes for 1 hour before and for 4 hours after administration, alongside routine clinical monitoring. This definition of hypotension was used to determine the dose-limiting toxicity of sildenafil in the study. Additionally, we recorded all concurrent therapies, such as the use of inotropes or vasopressors, hydrocortisone, sedatives, analgesics and antiseizure medications. The relationship between AEs and the study medication was initially assessed by the investigators at the time of the event and subsequently reviewed by the DMC.

Clinical Pharmacology

Plasma concentrations of sildenafil and its active main metabolite N-desmethyl sildenafil were quantified using a validated ultraperformance liquid chromatography method

coupled to tandem mass spectrometry, as previously described.[29] The lower limit of quantification for both analytes was 2.5 ng/mL. Accuracies (precision) ranged from −8.3% to +13.7% (3.8%-8.9%) for sildenafil, and −8.4% to +12.8% (3.0%-14.0%) for N-desmethyl sildenafil. Additionally, we quantified the downstream second messenger, cyclic guanosine monophosphate ,[30] using a commercially available Cyclic GMP ELISA Kit (Cayman Chemical), obtained (Biomol GmbH).[2]

Neuroimaging

Day-2 brain MRIs were performed during TH.[2,16,17,31] Neuroradiologists, blind to the neonates' clinical condition, reviewed the brain MRIs and reported the presence and extent of brain injury using a previously described MRI scoring system (BG – watershed score).[16,17,32,33] This system assigned a score of 0 for "no brain injury," 1 for "basal ganglia (BG) injury pattern," 2 for "watershed injury pattern," and 3-4 for "severe extensive brain injury," with a score of 4 indicating "near-total injury." Neonates with evidence of day-2 brain injury were eligible for the sildenafil treatment. As part of an exploratory safety analysis, we performed follow-up brain MRIs on the surviving neonates with HIE treated with TH at days 10 and 30 of life.[2,16,17,32] Reduction in brain injury was defined as decreased injuries in the BG, cortex or white matter, fewer cystic lesions, or reduced brain volume loss in the cortex and white matter.

Eighteen-Month Neurodevelopment

As part of our safety assessment, surviving neonates underwent structured neurodevelopmental evaluations at 18 months of age, conducted by pediatricians in the neonatal follow-up clinic or pediatric neurologists who were blinded to the neonates' clinical history and treatment group. Examinations documented the presence of cerebral palsy (CP), epilepsy, and global developmental delay (GDD).[32] GDD was defined as a significant, objectively measured delay in two or more developmental domains, including gross or fine motor function, speech and language, cognition, personal-social functioning, or activities of daily living, in accordance with the American Academy of Neurology and Child Neurology Society Practice Guidelines.[34] In addition, a psychologist performed comprehensive neurodevelopmental assessments using the Bayley Scales of Infant and Toddler Development, Third Edition to calculate motor, cognitive, and language composite scores.[35] Significant neurodevelopmental impairment was defined as a Bayley Scales of Infant and Toddler Development, Third Edition composite score <70, CP, epilepsy, or GDD.[2]

Statistical Analysis

All neonates with HIE treated with TH who received at least 1 dose of sildenafil were included in the analysis. AEs and SAEs were summarized with the proportion of neonates experiencing at least one event. Concentrations of sildenafil, N-desmethyl sildenafil, and cGMP over time were summarized using descriptive statistics by dose group. Neurodevelopmental outcomes at 18 months were summarized with the proportions of neonates with outcome of death or survival with significant neurodevelopmental impairment.

Results

Patients

Between October 2019 and December 2021, 119 neonates with HIE were treated with TH (**Figure 1**). Among them, 49 (41%) had moderate-severe HIE based on aEEG at admission. Twelve were ineligible for the study: 8 died on the first postnatal day, 2 had a genetic syndrome, and 2 were isolated because their mothers tested positive for COVID-19. Among the eligible neonates, parental consent rate was 81% (30/37). Of the 30 neonates enrolled for day-2 MRI, 20 showed signs of hypoxic-ischemic brain injury. Seven were excluded for the following reasons: intraventricular (n = 2) or intraparenchymal (n = 2) hemorrhage, underlying abnormal brain (brain hypoplasia, and diagnosed later with neuromuscular disorder) (n = 1), withdrawal of intensive care (n = 1), and parental refusal for the study drug (n = 1). The remaining 13 neonates were assigned to treatment, until at least 3 neonates in each group completed all 14/14 doses (8 in group 1 and 5 in group 2). The expansion phase in group 2 was stopped prematurely because of the COVID-19 pandemic. Neonates' characteristics are summarized in **Table I**. Baseline troponin I levels were 0.12 μg/L (IQR 0.07-0.47) in group 1 and 0.06 μg/L (IQR 0.05-2.22) in group 2. Baseline creatinine was 64.5 μmol/L (IQR 50.8-101.0) in group 1 and 133.0 μmol/L (IQR 98.0-184.0) in group 2.

Safety

Four neonates with near-total brain injury on day-2 (baseline) imaging and refractory seizures died following transition to palliative care—three neonates (38%) from group 1 and one (20%) from group 2. In group 1, two neonates died on postnatal day 3 after having received 1 and 2 doses of sildenafil, respectively. A third neonate was transitioned to palliative care on day 4 and died on day 11, having received four doses of sildenafil. One infant in group 1 was transitioned to palliative care on day 8, survived, and continued in the study. In group 2, one neonate was transitioned to palliative care on day 15 and died on day 25 after receiving 14 doses of sildenafil. All the other enrolled neonates were discharged alive.

All the enrolled infants, except for one in group 2, experienced seizures before starting sildenafil. As a result, they were receiving multiple antiseizure medications (phenobarbital, phenytoin, topiramate, levetiracetam, and/or midazolam) to manage their refractory seizures at baseline, before the study medication was initiated. In addition, 6/8

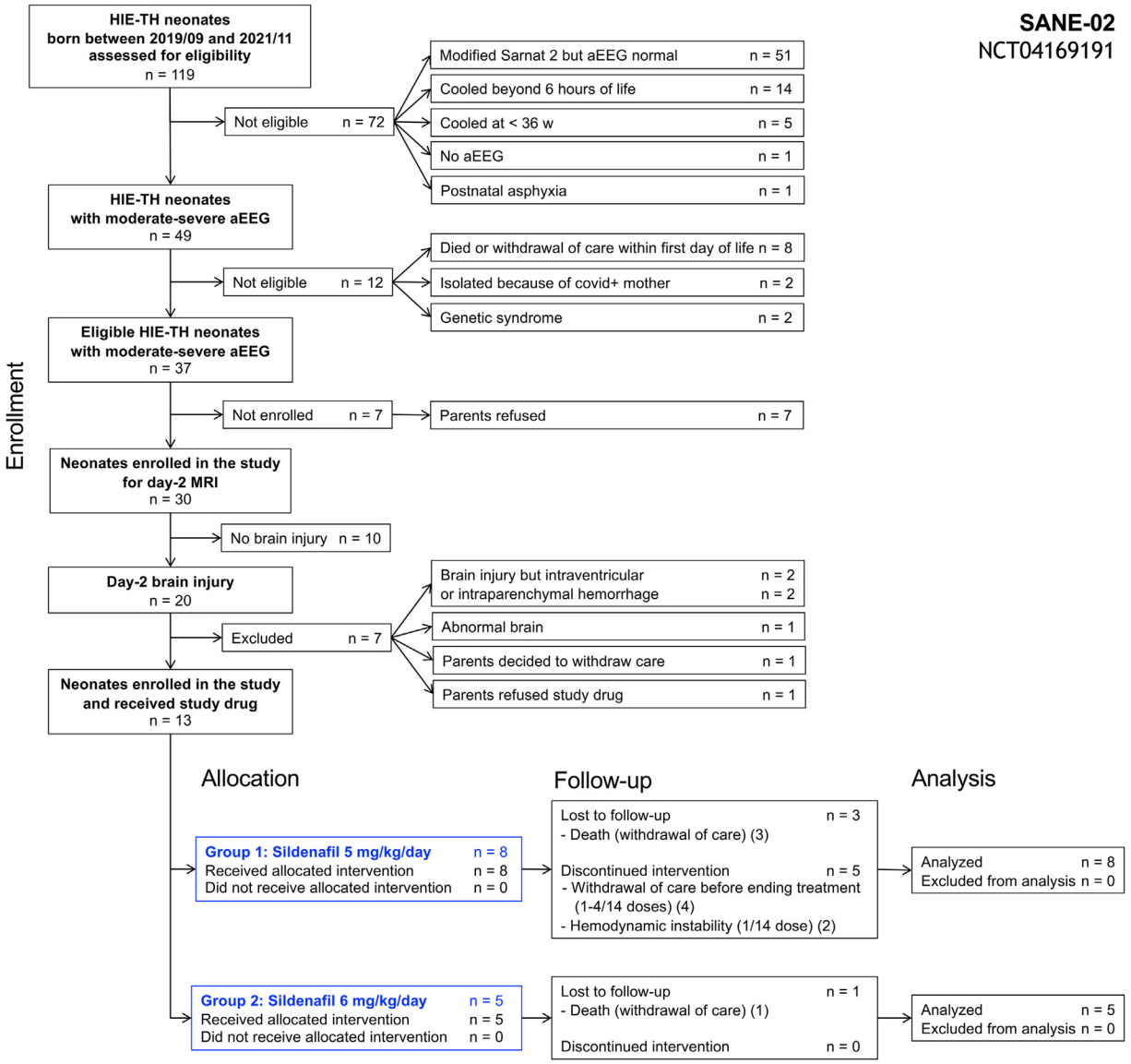


Figure 1. Flow chart of the participants in the SANE-02 trial. *aEEG*, amplitude-integrated electroencephalogram; *HIE*, hypoxic-ischemic encephalopathy; *MRI*, magnetic resonance imaging; *TH*, therapeutic hypothermia.

neonates in group 1 received inotropic or vasopressor agents and/or hydrocortisone before the first sildenafil dose; three met the pre-established discontinuation criteria after the first dose of sildenafil. In two, blood pressure dropped 15 minutes after the first dose of sildenafil; notably, both had received a loading dose of antiseizure medications (eg, phenobarbital, phenytoin, and/or midazolam) 1 hour prior to sildenafil administration. After consulting with the DMC, it was decided that for the next cohort, sildenafil doses would be delayed by 2 hours after an antiseizure medication administration. In group 2, all five neonates were on inotropic or vasopressor agents and/or hydrocortisone before the first sildenafil dose, and no significant blood pressure changes were observed following sildenafil administration.

No new intraventricular or intraparenchymal hemorrhages were found on follow-up MRIs, and there was no evidence of retinal vasculopathy in any of the neonates.

Pharmacological Studies

The average time from birth to the first sildenafil dose was 46.3 ± 5.0 hours. Six neonates received 3 doses and 3 received 4 doses of sildenafil during TH; all remaining doses were administered after rewarming. Figure 2 illustrates the concentration-time profiles of sildenafil and its active metabolite. After initiation during TH, sildenafil rapidly reached a steady-state concentration that persisted through the first 4 days of treatment before gradually declining due to an increasing metabolization capacity over the first week of life. In contrast, the N-desmethyl sildenafil levels steadily

Table I. Baseline characteristics of all the neonates with HIE treated with TH and sildenafil enrolled in SANE-02

Treatment	Group 1: Sildenafil up to 2.5 mg/kg/dose q12 h								Group 2: Sildenafil up to 3.0 mg/kg/dose q12 h				
Patients	**Gr.1 #1**	**Gr.1 #2**	**Gr.1 #3**	**Gr.1 #4**	**Gr.1 #5**	**Gr.1 #6**	**Gr.1 #7**	**Gr.1 #8**	**Gr.2 #1**	**Gr.2 #2**	**Gr.2 #3**	**Gr.2 #4**	**Gr.2 #5**
Gestational age, weeks	39.43	40.14	40.86	40.43	37.00	40.30	39.29	40.00	41.00	38.86	39.00	40.43	40.57
Birth weight, g	3460	5435	4210	3340	2540	2900	3265	3904	2610	3570	2740	3600	3320
Sex	Male	Female	Male	Male	Female	Male	Male	Male	Male	Male	Female	Male	Female
Delivery mode	Vaginal delivery	Vaginal delivery	Vaginal delivery	C-section	C-section	C-section	C-section	C-section	C-section	Vaginal delivery	C-section	C-section	C-section
10-min Apgar score	>5	0	5	5	6	5	6	6	>5	4	2	5	0
Cord pH	7.02	7.24	6.98	6.95	-	7.00	-	6.98	7.03	7.15	6.81	6.83	6.72
First gas pH	7.17	7.16	7.09	6.84	7.06	6.90	7.01	6.97	7.05	6.76	6.85	7.06	6.62
Lactate at admission, mmol/L	2.4	3.8	11.3	12.8	23.0	8.6	7.5	11.7	3.4	17.0	14.8	21.0	22.0
Modified Sarnat score on admission	2	3	2	2	3	2	3	2	2	2	2	2	3
aEEG on admission	Severe	Moderate	Moderate	Severe	Severe	Severe	Severe	Severe	Moderate	Moderate	Moderate	Severe	Severe
D-1 score for neonatal acute physiology II (SNAP-II)	39	15	39	55	41	51	39	48	39	27	51	46	75
Minimum d-1 PaO_2, mmHg	35.3	67.0	39.7	48.8	69.6	37.7	109.0	38.1	41.9	62.5	48.6	54.9	33.2
Nitric oxide during NICU hospitalization	No	No	No	Yes	No	Yes	No	Yes	No	No	No	No	Yes
D-1 vasoactive-inotropic score (VIS)[36]	5	0	0	20	6	7.5	0	6	5	8	16	5	5
Baseline* troponin I, μg/L	0.019	0.106	0.104	0.421	1.153	0.124	0.055	0.485	0.056	0.161	0.058	0.038	4.271
Baseline* creatine kinase, U/L	1342	8580	3392	8123	1982	4279	2482	1783	6915	7693	3207	2123	31 113
Baseline* creatinine, μmol/L	50	53	48	158	72	106	57	86	76	176	120	133	192
Maximum creatinine during NICU stay, μmol/L	79	72	67	266	72	110	86	93	97	176	124	222	662
Baseline* ALT, U/L	16	51	491	622	362	221	45	601	253	318	30	516	590
Maximum ALT during NICU stay, μmol/L	22	58	491	434	362	221	45	610	357	333	30	1237	590
Baseline* AST, U/L	44	198	293	1242	800	414	95	355	331	696	144	753	1346
Maximum AST during NICU stay, μmol/L	79	198	293	1242	800	719	95	437	745	696	144	821	1346
Baseline* CRP, mg/L	47.1	18.8	37.2	10.2	3.8	12.6	2.6	1.3	4.7	14.4	7.6	81.6	6.9
Baseline* WBC x10^9/L	17.5	17.5	6.3	17.3	16.4	8.8	13.4	14.9	4.2	10.8	18	10.7	21.6
Seizures	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Phenobarbital during NICU stay	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes
Phenytoin during NICU stay	No	No	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes
Midazolam during NICU stay	No	No	No	No	Yes	No	No	Yes	Yes	No	Yes	Yes	Yes
Morphine during TH	No	Yes	No	Yes	No	No	Yes	No	No	Yes	No	Yes	Yes
Fentanyl during TH	No	No	No	No	No	Yes	No	No	No	No	No	No	No
Dexmedetomidine during TH	No	No	No	No	No	No	No	No	No	No	No	No	No
Baseline* BG/W score	1	2	2	2	4	4	4	4	2	3	3	4	4
Time to first sildenafil dose, hours of life	46.0	40.7	46.2	34.9	50.0	50.0	49.2	45.4	52.3	45.8	45.6	41.8	52.7
Outcome	Alive	Alive	Alive	Alive	Alive	Dead†	Dead†	Dead†	Alive	Alive	Alive	Alive	Dead†
D-30 reduction of brain injury‡	Yes	Yes	Yes	Yes	No	-	-	-	Yes	Yes	Yes	No	-

ALT, alanine aminotransferase; *AST*, aspartate aminotransferase; *BG/W*, basal ganglia/watershed; *CRP*, C-reactive protein; *NICU*, neonatal intensive care unit; *PaO_2*, PO2 in arterial blood; *VIS*, vasoactive-inotropic score; *WBC*, white blood count.
Brain injury severity was scored using a previously described scoring system (BG/W score).[16,17,32,33]
*Baseline = on day 2 of life, before study medication administration.
†Died after parents opted for withdrawal of active care following quality-of-life considerations.
‡Reduction of brain injury included decreased injuries in basal ganglia, cortex or white matter, fewer cystic lesions or reduced brain volume loss in the cortex and white matter.

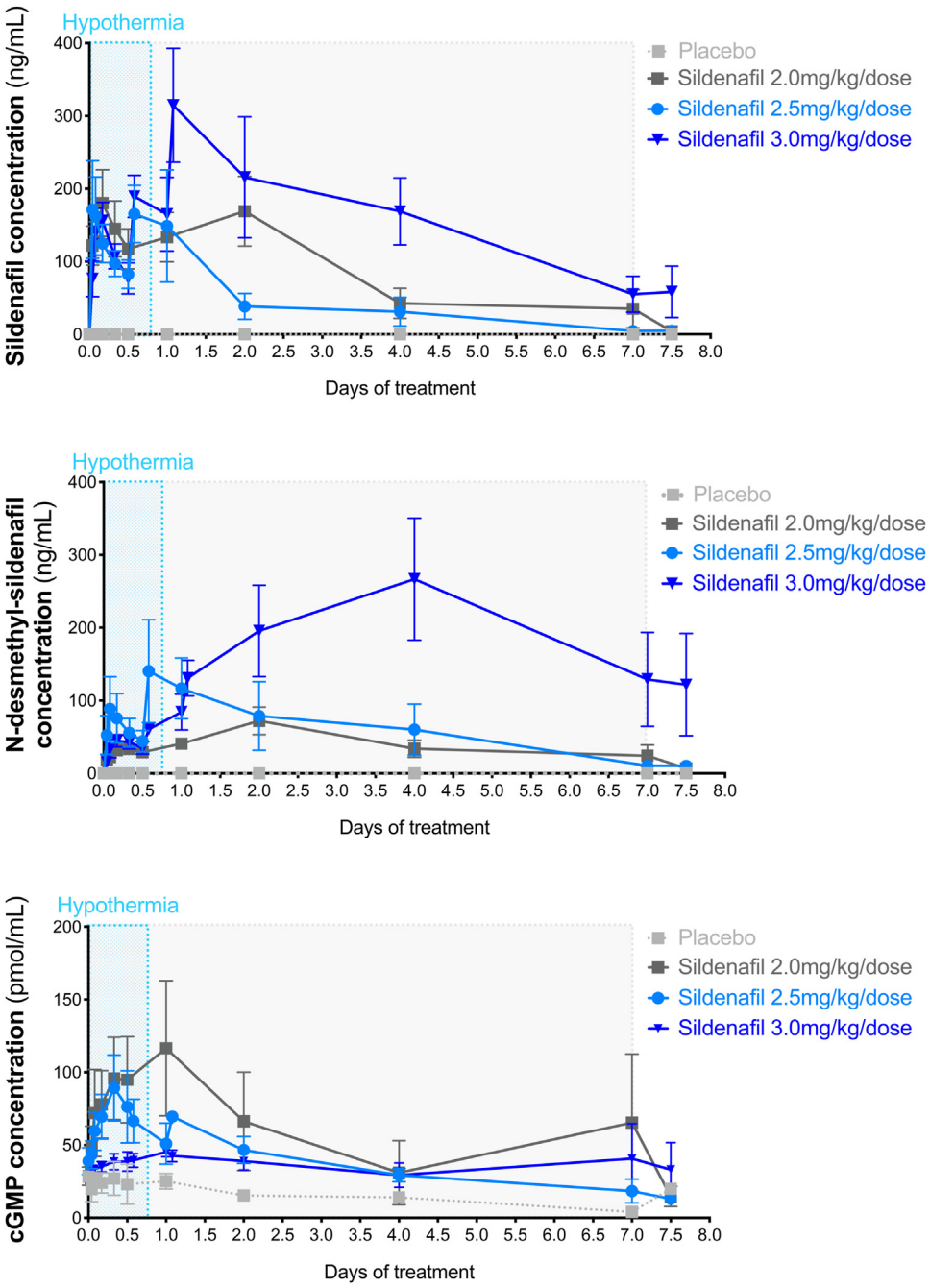


Figure 2. Concentrations of sildenafil, N-desmethyl sildenafil, and cGMP during treatment. Each enrolled infant received enteral sildenafil q12 h from day 2 to 3 of life for seven consecutive days, totaling 14 doses. Both groups began with a first dose of 2.0 mg/kg followed by a second dose of 2.5 mg/kg 12 hours later. Group 1 then received 2.5 mg/kg doses q12 h (5 mg/kg/day), whereas group 2 received 3.0 mg/kg doses q12 h (6 mg/kg/day). Representation of placebo group and sildenafil group from SANE-01 was kept for comparisons.[2] *cGMP*, cyclic guanosine monophosphate.

increased, peaking around day 4 of treatment, which also reflected an increasing metabolization. The median predicted average steady-state concentrations of sildenafil were 145 ng/mL following the initial 2.0 mg/kg dose, 181 ng/mL after increasing to the intermediate 2.5 mg/kg dose, and 215 ng/mL after escalating to the highest 3.0 mg/kg dose.

Neuroimaging

On the day-2 brain MRI, one participant in group 1 showed a BG injury pattern, three had a watershed injury pattern, and four had severe extensive brain injury. In group 2, one neonate had a BG injury pattern, and four had severe, extensive brain injury. The median timing of "day-10" MRIs was day 12 (range: 11-14) for group 1 and day 12 (range: 11-

Table II. Eighteen-month neurodevelopmental outcome of the neonates with hypoxic-ischemic encephalopathy treated with therapeutic hypothermia and sildenafil according to day-2 BG/W score

Treatment	Bayley-III (BSID-III) motor score	Bayley-III (BSID-III) cognitive score	Bayley-III (BSID-III) language score	CP	GDD	Epilepsy	Hearing loss requiring hearing aids	Significant neurodevelopmental impairment
Baseline* BG/W score 1								
Group 1, #1	88	100	74	No	No	No	No	No
Baseline* BG/W score 2								
Group 1, #2†	-	-	-	No	No	No	No	No
Group 1, #3	94	85	65	No	No	No	No	No
Group 1, #4	103	95	77	No	No	No	No	No
Group 2, #1	95	91	107	No	No	No	No	No
Baseline* BG/W score 3								
Group 2, #2	103	115	103	Yes (GMFCS 1)	No	No	No	Yes
Group 2, #3	91	85	74	No	No	No	No	No
Baseline* BG/W score 4								
Group 1, #5	73	80	68	No	Yes	No	Yes	Yes
Group 1, #6‡	-	-	-	-	-	-	-	-
Group 1, #7‡	-	-	-	-	-	-	-	-
Group 1, #8‡	-	-	-	-	-	-	-	-
Group 2, #4	55	47	46	Yes (GMFCS 4)	Yes	Yes	Yes	Yes
Group 2, #5‡	-	-	-	-	-	-	-	-

BG/W, basal ganglia/watershed score; *BSID-III*, Bayley Scales of Infant and Toddler Development, Third Edition; *GMFCS*, Gross Motor Function Classification System.
Eighty-nine percent (8/9) of the enrolled neonates, who survived, came back for the 18-month follow-up.
Brain injury severity was scored using a previously described scoring system (BG/W score).[16,17,32,33]
Significant neurodevelopmental impairment was defined as BSID-III <70, CP, GDD, and/or epilepsy.
*Baseline = on day 2 of life, before study drug administration.
†One neonate refused the Bayley-III assessment at the 18-month follow-up visit.
‡Four neonates did not survive the perinatal period.

13) for group 2. Day-10 brain MRIs were completed for all surviving participants through day 10 (10/10; 100%). For the "day-30" MRIs, the median timing was day 31 (range: 33-35) for group 1 and day 34 (range: 32-39) for group 2. Day-30 brain MRIs were obtained for all surviving participants through day 30 (9/9; 100%). Qualitatively, by the day-30 MRI, partial recovery of injury, fewer cystic lesions, and/or reduced brain volume loss (defined as widening of the supratentorial ventricular system) were observed in 80% (4/5) in group 1 and in 75% (3/4) in group 2, compared with the baseline day-2 brain MRIs.

Eighteen-Month Neurodevelopment

Eighty-nine percent (8/9) of surviving participants underwent an 18-month assessment, with a median age at follow-up of 19.5 months (IQR 18.1-20.8). The composite outcome of death or survival with significant neurodevelopmental impairment at 18 months was seen in 50% (4/8) in group 1 and 60% (3/5) in group 2. Among the survivors, 0% (0/5) in group 1 and 50% (2/4) in group 2 developed CP; 20% (1/5) in group 1 and 25% (1/4) in group 2 developed GDD; and 0% (0/5) in group 1 and 25% (1/4) in group 2 developed epilepsy (**Table II**).

Discussion

Because neonates with HIE treated with TH may have reduced drug clearance due to TH and liver impairment, to ensure the safety and tolerability of sildenafil treatment in neonates with HIE treated with TH, the SANE-01 study evaluated a medium-high dose of 2.0 mg/kg q12 h (4.0 mg/kg/day)[2,37,38] However, sildenafil concentration declined rapidly during the first postnatal week after TH was completed. Preclinical studies have shown that higher doses of sildenafil were more effective in repairing brain injury caused by HIE.[3,4] In our current SANE-02 study, we demonstrated that sildenafil was safe at doses up to 3.0 mg/kg q12 h (6.0 mg/kg/day) in neonates with HIE treated with TH, which is similar to the maximum recommended dose of sildenafil for treating neonatal pulmonary hypertension.[8,10,11]

In our trial, similarly to SANE-01, we deliberately enrolled the most critically ill neonates with HIE treated with TH—those with moderate or severe HIE on an aEEG at admission and confirmed day-2 brain injury while receiving TH. Our prior studies have shown that brain injury identified on day-2 brain MRI brain injury closely parallels post-TH injury in neonates with HIE treated only with TH.[16,17] Therefore, we anticipated that these neonates would have hemodynamic instability and active seizures at baseline.[12,14,24,27,39] As expected, most of our participants were receiving inotropic/vasopressor support at the time sildenafil was initiated. Our criteria for sildenafil discontinuation were pre-established, ensuring that the sildenafil was not stopped prematurely without standard treatments for these critically ill neonates. No significant blood pressure changes were observed, even at the highest dosage, when sildenafil administration was timed to avoid an overlap with antiseizure medications, which also can induce hypotension. Thus, a key consideration when using sildenafil in this population is to avoid administering it simultaneously with medications that can affect systemic vascular resistance, such as antiseizure drugs.

Enteral administration of sildenafil in neonates with HIE treated with TH led to plasma concentrations similar to those observed in normothermic infants born at term and

extremely preterm who were treated for pulmonary hypertension.[8,40,41] The values of the partially active metabolite N-desmethyl sildenafil also were concordant with concentrations found in neonates born prematurely.[41] This finding suggests that absorption and/or clearance of enteral sildenafil is not critically altered by gestational age, birth asphyxia, or TH. Sildenafil clearance rapidly increased in the first postnatal week, similar to the results of other trials reporting sildenafil pharmacokinetics.[10] Gradually reaching the maximum dose (3.0 mg/kg q12 h = 6.0 mg/kg/day) by the third dose likely improved safety and tolerability because by that time, neonates with HIE treated with TH were less critically ill and less prone to hypotension. At this dosage, sildenafil reached a steady-state concentration persisting after the cessation of TH, at a level that was not reached in the SANE-01 trial with the lower dose of 2.0 mg/kg q12 h (4.0 mg/kg/day). All the sildenafil doses resulted in average plasma concentrations high enough to ensure adequate exposure for a pharmacological effect.[8,40] This finding indicates that enteral sildenafil was effectively absorbed by neonates with HIE treated with TH.

Unlike the SANE-01 trial,[2] the SANE-02 trial did not include a placebo group. The primary objective of this phase 1b study was to evaluate the tolerability and safety of higher sildenafil doses, while avoiding the collection of additional blood samples from the neonates with HIE treated with TH receiving a placebo. Previously, we reported that none (0/3; 0%) of the placebo-exposed neonates in SANE-01 showed a reduction of brain injuries on a day-30 MRI,[2] which was consistent with earlier studies that performed repeat neuroimaging in similar neonates.[16,17] In contrast, 71% (5/7) of neonates in the SANE-01 sildenafil group and 78% (7/9) in SANE-02 exhibited a reduction in brain injury. Although these results are descriptive and exploratory, they indicate a favorable safety profile and a promising trend, and support proceeding to further investigation of efficacy in a multicenter randomized phase 2 and 3 trials. Furthermore, the composite outcome of mortality or significant neurodevelopmental impairment at 18 months was observed in 100% (3/3) of the SANE-01 placebo group, 57% (4/7) of the SANE-01 sildenafil group, and 54% (7/13) of the SANE-02 sildenafil groups. Among the surviving neonates, the incidence of CP was 100% (3/3) in the SANE-01 placebo group, 17% (1/6) in the SANE-01 sildenafil group, and 22% (2/9) in the SANE-02 sildenafil groups. Although the sample size of both SANE trials was small, it is important to highlight that many neonates with HIE treated with TH treated with sildenafil in both SANE-01 and SANE-02 were critically ill at baseline. Many of them had near-total brain injury on day-2 MRI, elevated troponin levels (reflecting impaired cardiac function), and high creatinine values (indicating impaired kidney function). We intentionally enrolled the highest-risk neonates in our trials, in contrast to other trials testing neuroprotective therapies in neonates,[42] which recruited a broad range of HIE severity, including neonates without brain injury. We did not anticipate a reduction in mortality rates, since the study drug was initiated only after brain injury was confirmed, and many parents opted for palliative care when faced with confirmed brain injury.[43] Although our descriptive findings on neurological and neurodevelopmental safety are promising, they need validation by a phase 2 trial, with a larger cohort of neonates with HIE treated with TH and an expanded placebo group. This trial will enable us to assess multicenter feasibility and safety and to further explore sildenafil's potential efficacy in repairing brain injury in neonates with HIE treated with TH.

Based on the results of the SANE-01 trial, which demonstrated a good enteral absorption of sildenafil, we chose the enteral route for SANE-02 as well. This is the most commonly used route for administering sildenafil to neonates worldwide, offering the same efficacy as the intravenous route, but with fewer side effects.[44] In addition, the intravenous form is not universally available, is more expensive, and requires specialized skills for proper administration, which makes it less practical for widespread use.[45] Although the enteral absorption of sildenafil can vary,[8,30,46] its beneficial effect has been shown in neonates with persistent pulmonary hypertension.[9,36,47] The recommended neonatal dose range of sildenafil is 0.5-3.0 mg/kg/dose, up to a maximum of 6.0 mg/kg/day. Although sildenafil for pulmonary hypertension typically begins at low dose and is gradually increased, our preclinical data showed greater brain repair at higher doses. Based on these findings, we initiated treatment at the higher end of the dosing range to maximize neurorestorative potential from the outset.

The day-30 brain MRI served as a short-term neurodevelopmental proxy, since previous studies have shown that it correlates with the day-10 MRIs performed in neonates with HIE treated with TH alone and with their neurodevelopment at 18 months.[16,17,32] Most neonates had been discharged home by this time and underwent their MRIs without sedation, using the wrapping and feeding method.[31] High-quality images were successfully obtained for all the surviving neonates.

Our study is limited by its single-center design, small sample size, and recruitment challenges due to the ongoing COVID-19 pandemic. Our data are applicable only to the highest-risk neonates with confirmed brain injury and should not be extrapolated to broader HIE populations, particularly those with milder forms of the condition. However, our study provides valuable data to inform a future randomized trial, studying sildenafil at a therapeutic dosage that would provide maximal efficacy.

In conclusion, enteral sildenafil at doses up to 3.0 mg/kg/dose, given twice daily, was found to be safe and well tolerated in a small sample of neonates with HIE treated with TH enrolled in a single center. However, further testing in larger multicenter phase 2 and 3 trials is needed to assess the feasibility of this treatment strategy across multiples sites and its efficacy. ■

CRediT authorship contribution statement

Pia Wintermark: Writing – original draft, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Anie Lapointe:** Writing – review

& editing, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Gabriel Altit:** Writing – review & editing, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Robin Steinhorn:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Emmanouil Rampakakis:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Andreas D. Meid:** Writing – review & editing, Formal analysis, Data curation. **Jürgen Burhenne:** Writing – review & editing, Methodology, Investigation. **Gzona Bajraktari-Sylejmani:** Writing – review & editing, Investigation, Formal analysis. **May Khairy:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Marie-Therese Adamo:** Writing – review & editing, Methodology, Investigation. **Guillaume Gilbert:** Methodology, Data curation. **Daniela Toffoli:** Writing – review & editing, Methodology, Investigation. **Samara Zavalkoff:** Writing – review & editing, Supervision, Investigation. **Thuy Mai Luu:** Writing – review & editing, Supervision, Investigation. **Elizabeth Hailu:** Writing – review & editing, Investigation. **Walter E. Haefeli:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

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Data Statement

Data sharing statement available at www.jpeds.com.

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